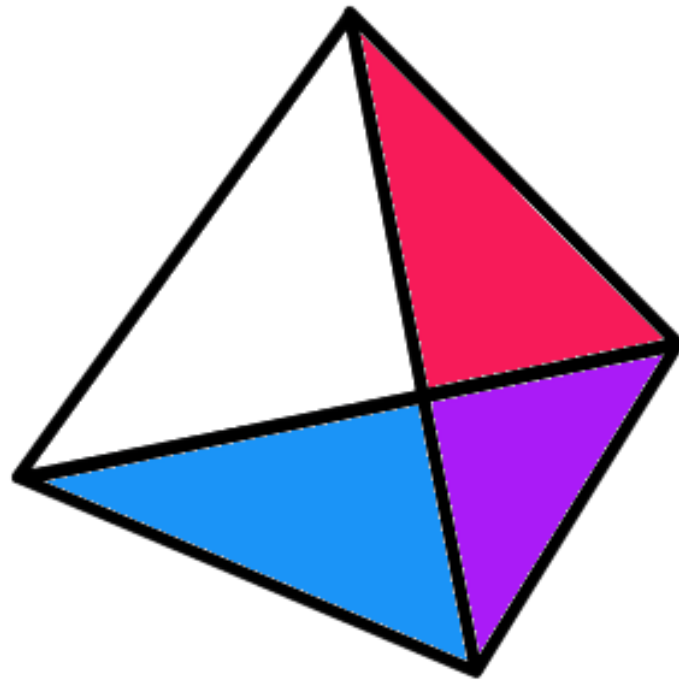




Project PRISM

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1. Linear Kinematics

1.1 Displacement

Displacement is a vector quantity which represents the net change in an object's position. The SI unit for displacement is m (Metres).

The symbol for displacement is s .

1.2 Velocity

Velocity is a vector quantity which represents the rate of change of an object's displacement with respect to time. The SI unit for velocity is m s^{-1} (Metres per second).

The symbol for velocity is v .

u may be used to denote the initial velocity of an object, in which case v denotes the final velocity of the object.

Mathematically, velocity is the time derivative of displacement, $v = \frac{ds}{dt}$.

1.3 Acceleration

Acceleration is a vector quantity which represents the rate of change of an object's velocity with respect to time. The SI unit for acceleration is m s^{-2} (Metres per second squared).

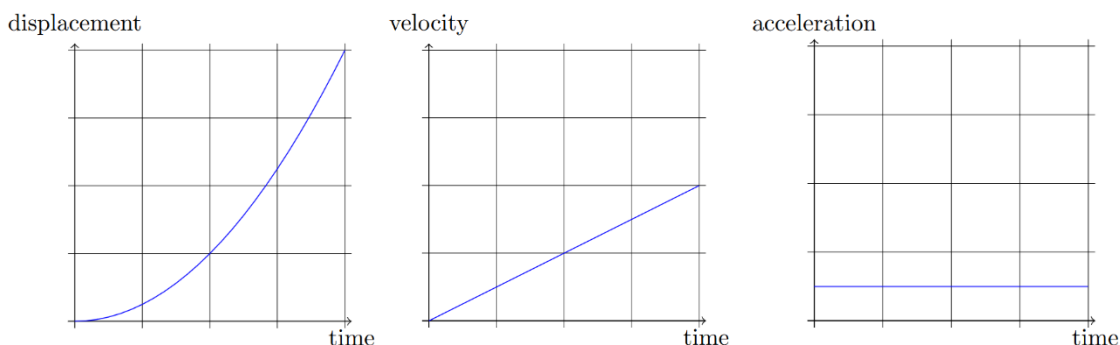
The symbol for acceleration is a .

Mathematically, acceleration is the time derivative of velocity, $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$.

The acceleration due to gravity on Earth is 9.81 m s^{-2} .

1.4 Graphs of Displacement, Velocity, & Acceleration

These equations can be derived graphically or algebraically.



The above graphs all represent the same motion of a particle object undergoing constant acceleration from rest.

Note that the value of velocity is equal the area under the acceleration-time graph; and that the value of displacement is equal to the area under the velocity-time graph.

1.5 SUVAT Equations

There is a set of equations which governs the behaviour of objects experiencing uniform acceleration. In these equations, the following symbols are used:

- s : Displacement
- u : Initial velocity
- v : Final velocity
- a : Acceleration
- t : Time

It is important to ensure that the base units used for each quantity are consistent.

1. $s = ut + \frac{at^2}{2}$
2. $s = vt - \frac{at^2}{2}$
3. $s = \frac{(u+v)t}{2}$
4. $v = u + at$
5. $v^2 = u^2 + 2as$

1.6 Kinetic Energy

The translational kinetic energy of an object is given by $KE = \frac{1}{2}mv^2$.

The SI unit for kinetic energy is J (Joules). 1 J is equal to $1 \text{ kg m}^2 \text{ s}^{-2}$.

2. Forces, Momentum, & Impulse

2.1 Momentum

Momentum is a vector quantity which is the product of an object's velocity and mass. The SI unit for momentum is kg m s^{-1} (kilogramme-metres per second).

The symbol for momentum is p .

Mathematically, $p = mv$.

2.2 Impulse

Impulse is a vector quantity which represents the change in the momentum of an object. Impulse can also be interpreted as the integral of force with respect to time. It uses the same units as momentum.

The symbol for impulse is Δp . (Δ indicates a change in a value)

Mathematically, $\Delta p = \int F \, dt$.

2.3 Force

Force is a vector quantity which represents the rate of change of momentum of an object. The SI unit for force is N (Newtons). 1 N is equal to $1 \, \text{kg m s}^{-2}$.

The symbol for force is F .

Mathematically, force is the time derivative of momentum, $F = \frac{dp}{dt}$.

When mass is constant, $F = \frac{d(mv)}{dt} = m \frac{dv}{dt} = ma$.

When velocity is constant, $F = \frac{d(mv)}{dt} = v \frac{dm}{dt}$.

2.4 Momentum & Impulse Equations

- $v = \frac{p}{m}$
- $\Delta v = \frac{\Delta p}{m}$
- $\Delta p = F\Delta t$

2.5 Newton's 2nd Law

Newton's 2nd Law states, for an object with constant mass:

$$F_{net} = ma$$

Where:

- F_{net} is the net force on the object
- m is the mass of the object
- a is the acceleration of the object

2.6 Newton's 3rd Law

Newton's 3rd Law states that for every action, there is an equal and opposite reaction. Consider 2 objects, A and B . For every force that A imparts on B , B imparts a force on A with equal magnitude and opposite direction.

It can be said that forces exist in pairs.

For example, the gravitational force the Earth exerts on a person forms a force pair with the gravitational force the person exerts on the Earth. However, since the mass of the Earth is so large, the Earth experiences negligible acceleration due to the gravitational pull of the person.

2.7 Conservation of Linear Momentum

The principle of conservation of linear momentum states that the total momentum of an isolated system (where there are no external forces) remains constant. This can be derived from Newton's 3rd Law.

Consider an isolated system with 2 objects — A and B . We first define the total momentum of the system as:

$$p = p_A + p_B$$

By the definition of an isolated system, the only force pair that can exist must be between A and B . We denote the force on A as F_A and the force on B as F_B . Hence, from Newton's 3rd Law, we can see that:

$$\begin{aligned} F_A &= -F_B \\ \frac{dp_A}{dt} &= -\frac{dp_B}{dt} \end{aligned}$$

We can find the rate of change of the total momentum of the system,

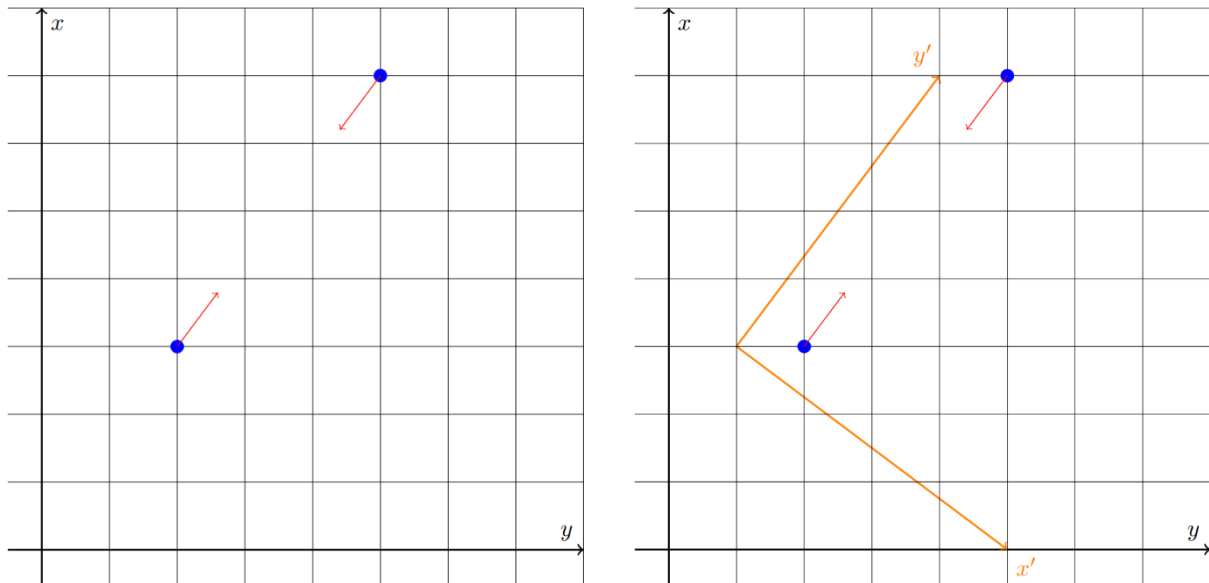
$$\begin{aligned} \frac{dp}{dt} &= \frac{dp_A}{dt} + \frac{dp_B}{dt} \\ \frac{dp}{dt} &= -\frac{dp_B}{dt} + \frac{dp_B}{dt} \\ \frac{dp}{dt} &= 0 \end{aligned}$$

Since the rate of change of p is zero, we can conclude that the total momentum of the system is constant.

This can be generalised to isolated systems with any number of objects.

2.8 Momentum in 2 Dimensions

To perform calculations regarding momentum in 2 dimensions, we can consider the momentum of objects as the sum of 2 momentum vectors along the x and y axes. Calculations for momentum in the 2 axes can be reduced to a single axis. This is done by redefining the coordinate axes such that the collision only affects the velocities of the object about 1 axis. In other words, one axis is parallel to the line of impact and one axis is perpendicular to the line of impact.



In the above diagram, the velocities of the objects are at a 53° angle to the x -axis. We can define our new axes (x' and y') such that they are also at a 53° angle to the old axes.

The velocities of the objects are parallel to the y' -axis and perpendicular to the x' -axis. Hence, the collision can be treated as a 1-dimensional collision along the y' -axis. In this example, the objects have no velocity component along the x' -axis. If the objects had velocity components along the x' -axis, they would remain unchanged as the change in momentum due to the collision is only along the y' -axis.

3. Rockets

Questions involving rockets sometimes take the form of:

A rocket has an initial mass of m_0 kg, it consumes fuel at a constant rate of $x \text{ kg s}^{-1}$, its exhaust velocity is $v_e \text{ m s}^{-1}$.

Thrust equation (Note that this is NOT the same as net force):

$$F_{\text{Thrust}} = xv_e$$

Tsiolkovsky Rocket Equation (Note that this does NOT account for gravity):

$$v = v_e \ln \frac{m_0}{m_f}$$

Where m_f is the final mass of the rocket.

To account for the effect of gravity on the motion of the rocket, the above equation can be extended to:

$$v = v_e \ln \frac{m_0}{m_f} - gt$$

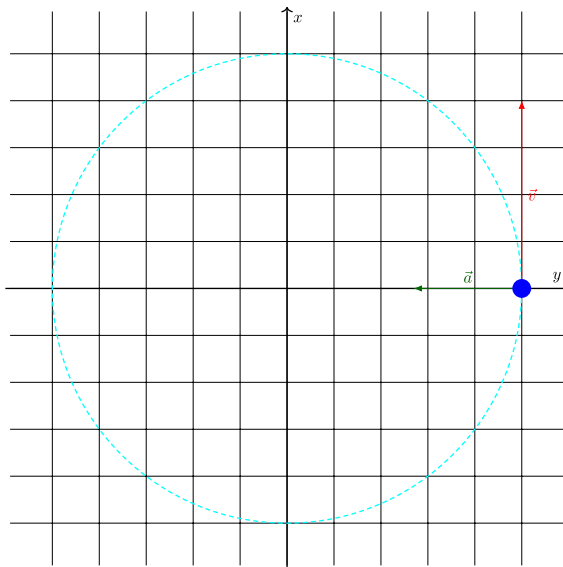
Where t is the burn time of the rocket.

4. Centripetal Acceleration

For a point mass in motion which fulfils the following criteria:

- The point mass is moving in a circular path about a central pivot
- The speed of the point mass is constant

It experiences an acceleration known as centripetal acceleration. Centripetal acceleration is a 'centre-seeking' acceleration, meaning that the direction of the acceleration vector is always pointed at the centre of the object's path.



The magnitude of centripetal acceleration is:

$$a_c = \frac{v^2}{r}$$

Where:

- a_c is the magnitude of centripetal acceleration
- v is the velocity of the point mass
- r is the radius of the object's path

From Newton's 2nd Law, the magnitude of centripetal force, F_c , is $m \frac{v^2}{r}$. The origin of centripetal force may vary (e.g. tension in a string, gravity, Lorentz force).

5. Moments

A moment, also known as torque, refers to the turning effect of a force about a pivot.

The moment due to a force about a pivot can be written as:

$$\tau = r_{\perp} F$$

Where:

- τ is the magnitude of the moment due to the force about the pivot
- $r_{\perp}$ is the perpendicular distance from the line of action of the force to the pivot

- F is the magnitude of the force

The line of action of a force refers to an imaginary straight line passing through the origin of the force vector with the same direction as the force vector.

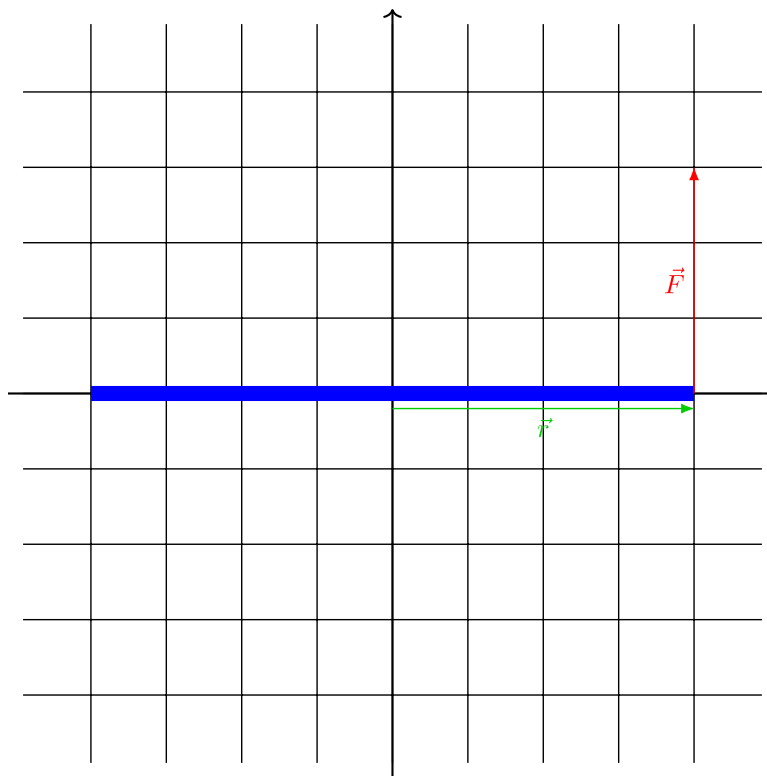
In terms of vectors, the equation can be written as:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Where:

- $\vec{r}$ is the displacement vector from the pivot to the origin of the force vector
- $\vec{F}$ is the force vector

The SI unit for moment is N m (Newton-metres).



Notice the since a cross product is used, the torque vector is directed perpendicularly to both the displacement and force vector. The direction of the torque vector indicates the axis about which the torque vector is acting. To find the direction of turning caused by the torque, the right-hand curl rule can be used.

Moments will be further explored later under the topic of rotational kinematics.

6. Equilibrium

6.1 Conditions for Equilibrium

An object in equilibrium does not experience linear or rotational acceleration. For an object to be in equilibrium, two conditions must be met:

- The net force on the object must be zero, this is equivalent to Newton's 1st Law
- The net moment on the object must be zero, this is known as the principle of moments.

6.2 Centre of Mass Calculations

The centre of mass of an object refers to a point that the mass of the object can be interpreted as acting through. This is useful when calculating moments (e.g. taking the centre of mass as a pivot point).

For a system of n discrete point masses, the centre of mass of the object can be calculated as:

$$\overrightarrow{r_{CoM}} = \frac{\sum_{i=1}^n \vec{r}_i m_i}{\sum_{i=1}^n m_i}$$

Where:

- $\vec{r}_i$ is the distance vector from the origin to the i -th particle
- m_i is the mass of the i -th particle

For solid objects, the object must be broken down into differential mass elements:

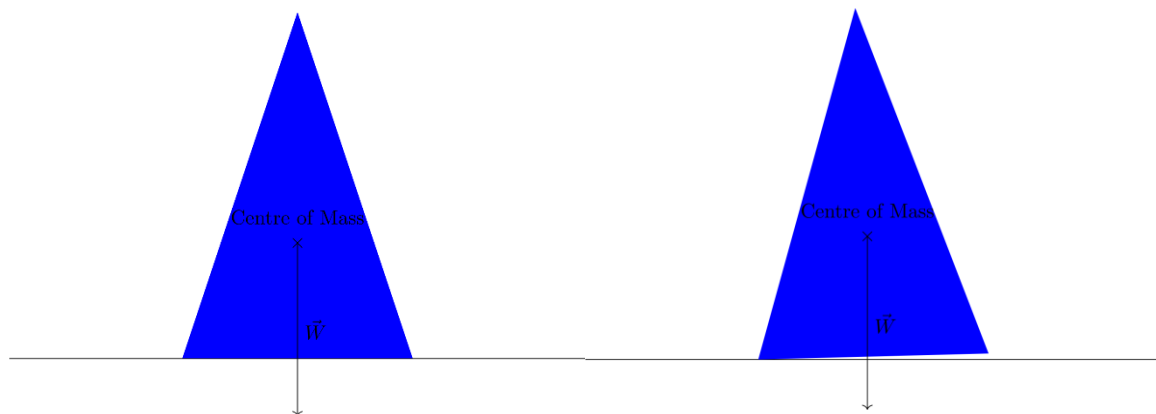
$$\overrightarrow{r_{CoM}} = \frac{\int \vec{r} dm}{m}$$

Where:

- $\overrightarrow{r_{CoM}}$ is a vector from the origin to the centre of mass
- $\vec{r}$ is a vector from the origin to a differential mass element dm
- m is the total mass of the object

6.3 Stable, Unstable, & Neutral Equilibrium

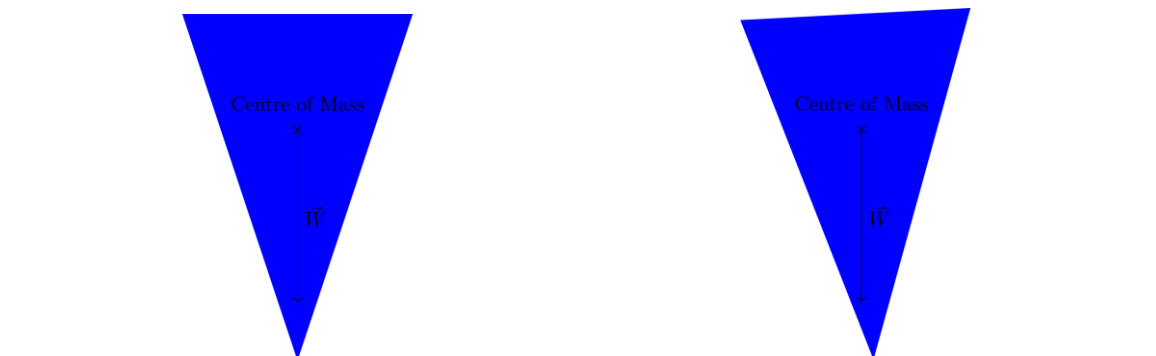
6.3.1. Stable Equilibrium



When the object experiences a minor displacement, the moment due to the weight of the object causes it to return to its position of stable equilibrium.

The line of action of weight is within the base of the object.

6.3.2. Unstable Equilibrium



When the object experiences a minor displacement, the moment due to the weight of the object causes its displacement to increase.

The line of action of weight is outside the base of the object.

6.3.3. Neutral Equilibrium



When the object experiences a minor displacement, the moment due to the weight of the object is zero.

The line of action of weight is directly over the support point of the object.

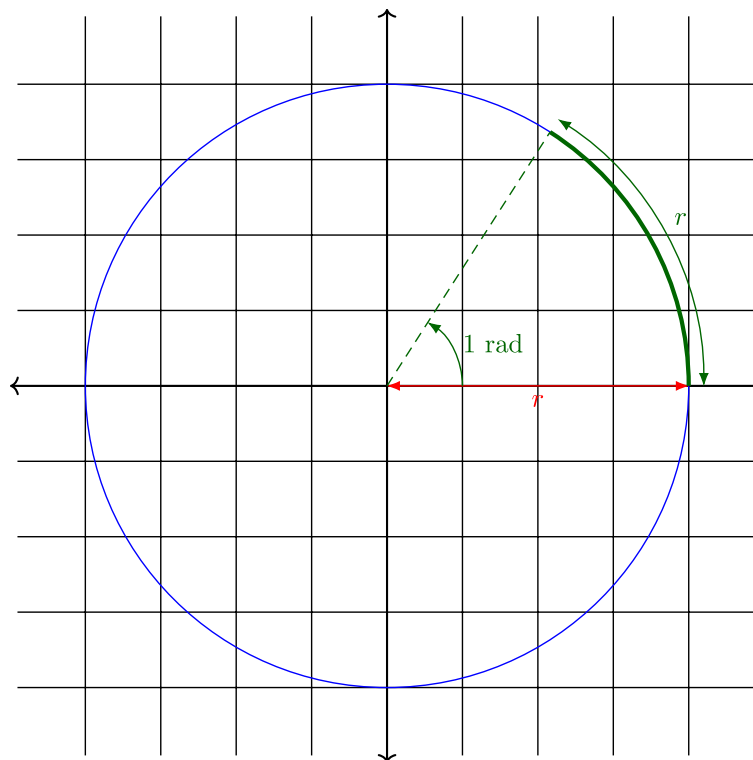
In the diagram, the orange line indicates the initial angular orientation of the disc and the orange cross indicates the initial point of contact between the disc and the ground; The red line and red cross indicate the final orientation and point of contact, respectively, after an angular displacement of 3° clockwise.

7. Rotational Kinematics

7.1 Radian Measure

To describe rotations, two units are commonly used, $^\circ$ (degrees) and radians. Radians are more commonly used in physics as they are more convenient to work with.

1 radian is defined as the angle at the centre of a circle subtended by an arc of length equal to the radius of the circle. This can be represented by the following diagram:



There are 2π radians in a full circle.

Note that radians are technically a dimensionless quantity as they represent a ratio of lengths. However, the unit 'rad' or ' c ' may sometimes be used to denote that angles are measured in radians. When no units are given, it is best to assume that radian measure is used.

7.2 Basic Equations

For a point mass rotating about a pivot, the following symbols and equations are used to relate its linear and rotational kinematic values. Radian measure is used.

- s : Distance travelled
- v : Tangential velocity
- a : Tangential acceleration
- θ : Angular displacement
- ω : Angular velocity (This is a lowercase omega, NOT a w)
- α : Angular acceleration
- r : Radius of motion

Equations:

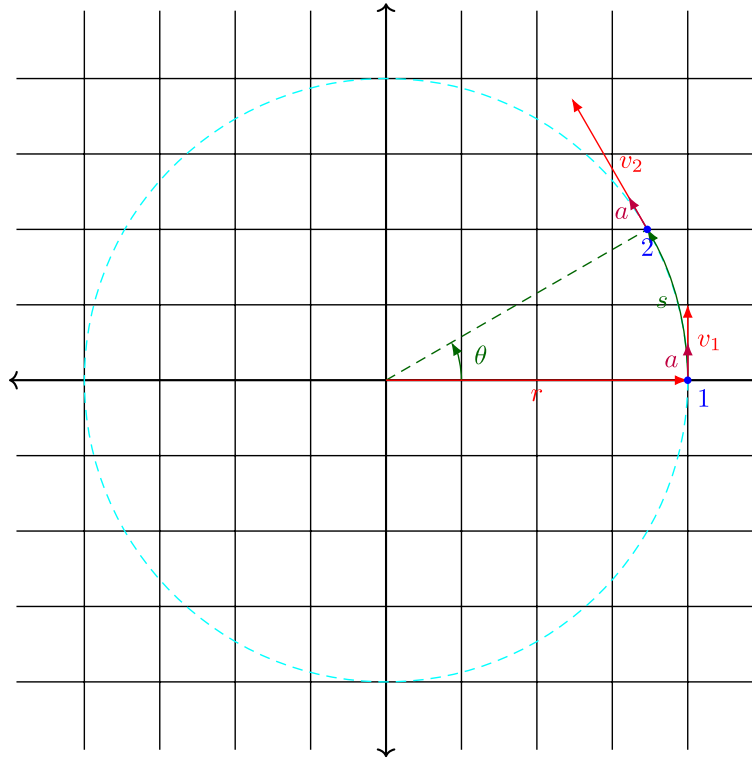
1. $s = \theta r$
2. $v = \omega r$
3. $a = \alpha r$

The basic equations for rotational kinematics, assuming uniform constant acceleration, are very similar to the SUVAT equations. In these equations, the following symbols are used:

- θ : Angular displacement
- ω_1 : Initial angular velocity
- ω_2 : Final angular velocity
- α : Angular acceleration
- t : Time

It is important to ensure that the base units used for each quantity are consistent.

1. $\theta = \omega_1 t + \frac{\alpha t^2}{2}$
2. $\theta = \omega_2 t - \frac{\alpha t^2}{2}$
3. $\theta = \frac{(\omega_1 + \omega_2)t}{2}$
4. $\omega_2 = \omega_1 + \alpha t$
5. $\omega_2^2 = \omega_1^2 + 2\alpha s$



This diagram shows the movement of a point mass rotating about the origin from position 1 to position 2. The mass experiences constant angular acceleration. The linear velocity and tangential acceleration vectors are shown. The distance travelled by the point mass is also shown.

Note that in this diagram, the vectors $\vec{\omega}_1$, $\vec{\omega}_2$, and $\vec{\alpha}$ are not shown. If they were shown, they would be vectors pointing out of the page at the origin. The direction of the angular velocity and angular acceleration vectors can be found by the right-hand curl rule.

Note that angular velocities and angular accelerations can be represented as vectors as they obey vector addition. However, angular displacements are not represented as vectors as they do not obey vector addition (e.g. non-commutative addition).

7.3 Moment of Inertia

We can establish an equivalent for Newton's 2nd Law for rotational kinematics. The following derivation is used to relate the angular acceleration, α , of a point mass to the torque, τ , acting upon it due to a tangential force, F , a distance r from its pivot:

$$F = ma$$

$$F = m\alpha r$$

$$Fr = m\alpha r^2$$

$$\tau = mr^2\alpha$$

Comparing this equation to Newton's 2nd Law ($F = ma$), we can see that the 'rotational equivalent' of mass is mr^2 . This is known as the moment of inertia, given the symbol I and the unit kg m^2

The formula $I = mr^2$ only applies to point masses. To calculate the moment of inertia of a solid object, it must first be broken down into differential mass elements. The moment of inertia about a specific axis of rotation is then given by:

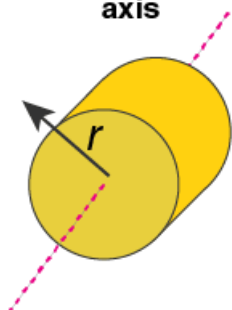
$$I = \int r^2 dm$$

Where:

- I is the moment of inertia of the object
- r is the distance between the differential mass element dm and the axis of rotation

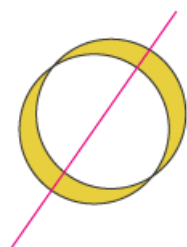


Solid cylinder or disc, symmetry axis



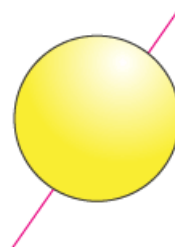
$$I = \frac{1}{2} MR^2$$

Hoop about symmetry axis



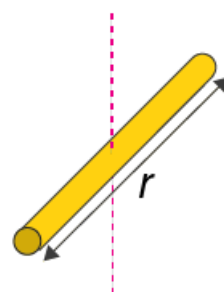
$$I = MR^2$$

Solid sphere



$$I = \frac{2}{5} MR^2$$

Rod about center



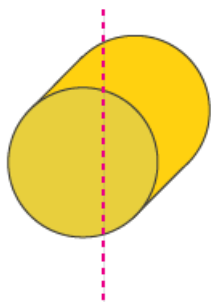
$$I = \frac{1}{12} ML^2$$

$$I = \frac{1}{4} MR^2 + \frac{1}{12} ML^2$$

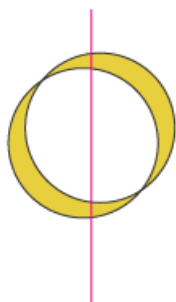
$$I = \frac{1}{2} MR^2$$

$$I = \frac{2}{3} MR^2$$

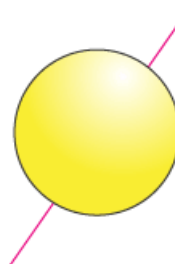
$$I = \frac{1}{3} ML^2$$



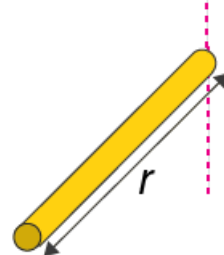
Solid cylinder central diameter



Hoop about diameter



Thin spherical shell



Rod about end

7.4 Conservation of Angular Momentum

For an object rotating about an external pivot, its angular momentum about an axis is defined as:

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

Where:

- $\vec{L}$ is the angular momentum of the object
- $\vec{r}$ is the distance vector from a chosen axis to the object's centre of mass
- $\vec{p}$ is the linear momentum vector of the object
- $\vec{v}$ is the velocity vector of the object
- m is the mass of the object

This quantity is also known as orbit angular momentum.

For an object rotating about its centre of mass, its angular momentum is defined as:

$$\vec{L} = I\vec{\omega}$$

- I is the moment of inertia of the object about an axis that is parallel to $\vec{\omega}$ and passes through its centre of mass (i.e. the moment of inertia about its centre of mass)
- $\vec{\omega}$ is the angular velocity of the object

This quantity is also known as spin angular momentum.

The total momentum of an object is the sum of its spin angular momentum and orbit angular momentum:

$$\vec{L} = \vec{r} \times m\vec{v} + I\vec{\omega}$$

7.5 Parallel Axis Theorem

The parallel axis theorem defines a relationship between the moment of inertia about an axis through an object's centre of mass to the moment of inertia about any parallel axis:

$$I = I_{CoM} + mr^2$$

Where:

- I is the moment of inertia about the new axis
- I_{CoM} is the moment of inertia about the axis through the centre of mass
- r is the distance between the two axes

8. Coriolis force

The Coriolis force refers to an inertial force that appears to act on objects when viewed from a rotating frame of motion.

9. Energy

The SI unit for energy is J (Joules). 1 J is equal to $1 \text{ kg m}^2 \text{ s}^{-2}$. Energy is a scalar quantity.

Work done can be defined as the amount of energy transferred from one object to another.

A common definition of work done is:

$$W = \vec{F} \cdot \vec{s}$$

or

$$W = \int \vec{F} \cdot d\vec{s}$$

Where:

- W is the work done on an object
- $\vec{F}$ is the force on an object
- $\vec{s}$ is the displacement of the object as it experiences the force

9.1 Law of Conservation of Energy

The law of conservation of energy states that energy cannot be created or destroyed, it can only be converted from one form to another. The total energy within an isolated system remains constant.

Some forms of energy are:

- Kinetic energy
 - Translational
 - Rotational
- Thermal energy
- Gravitational potential energy
- Elastic potential energy

9.2 Kinetic Energy

The kinetic energy of a point mass is given by:

$$KE = \frac{1}{2}mv^2$$

Where:

- KE is the kinetic energy of the point mass
- m is the mass of the point mass
- v is the velocity of the point mass

This formula can be derived from the definition of work done.

9.2.1. Translational Kinetic Energy

During translation, all parts of an object are moving at the same speed. Hence, the formula for translational kinetic energy is the same as the formula for the kinetic energy of a point mass:

$$KE = \frac{1}{2}mv^2$$

9.2.2. Rotational Kinetic Energy

During rotation, different parts of an object move at different speeds. The speed of a differential mass element dm is given by:

$$v = \omega r$$

Where:

- v is the velocity of dm
- ω is the rotational velocity of the object
- r is the distance of dm from the axis of rotation

Hence, the differential kinetic energy dKE is given by:

$$dKE = \frac{1}{2}(dm)\omega^2 r^2 = \frac{1}{2}(dm)r^2 \omega^2$$

We can find the total kinetic energy by integrating dKE with respect to dm :

$$\begin{aligned} KE &= \int \frac{1}{2}r^2 \omega^2 dm \\ &= \frac{1}{2}\omega^2 \int r^2 dm \end{aligned}$$

Earlier we derived the moment of inertia of an object as $\int r^2 dm$.

$$KE = \frac{1}{2}I\omega^2$$

Where:

- KE is the kinetic energy of the object
- I is the moment of inertia of the object about the axis of rotation
- ω^2 is the rotational velocity of the object

9.3 Thermal Energy

The change in the thermal energy of a substance is directly proportional to the change in its temperature:

$$Q = mc\Delta\theta$$

Where:

- Q is the change in thermal energy
- m is the mass of the substance
- c is the specific heating capacity of the substance (unit: $\text{J K}^{-1} \text{g}^{-1}$ or $\text{J }^{\circ}\text{C}^{-1} \text{g}^{-1}$)
- $\Delta\theta$ is the change in the temperature of the substance (unit: K or $^{\circ}\text{C}$)

9.4 Gravitational Potential Energy

For situations where the gravitational field strength can be approximated as a constant value regardless of location (e.g. near the surface of the Earth), the weight of the object is constant. Hence, the GPE formula is a simple $\vec{F} \cdot \vec{s}$ equation.

$$GPE = mgh$$

Where:

- GPE is the gravitational potential energy of the object
- m is the mass of the object
- g is the gravitational field strength
- h is the height of the object

9.5 Elastic Potential Energy

The EPE of an object is given by:

$$EPE = \frac{1}{2}kx^2$$

Where:

- EPE is the elastic potential energy of the object
- k is the spring constant (unit: N m^{-1})
- x is displacement

This topic will be covered more in-depth in later notes.

10. Collisions

There are 3 main types of collisions:

- Elastic: The kinetic energy of the system is conserved
- Partially inelastic: Some kinetic energy is lost
- Completely inelastic: Some (more than for a partially elastic collisions) kinetic energy is lost; Objects stick together after the collision.

In all cases, the linear momentum of the system is still conserved.

10.1.1. Elastic Collisions

Consider two objects which are going to collide, denoted A and B .

From the conservation of linear momentum,

$$p_{A,initial} + p_{B,initial} = p_{A,final} + p_{B,final}$$

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

11. Fluid Dynamics

11.1 Pressure

Pressure is defined as force per unit area. The SI unit for pressure is Pa (Pascals). 1 Pa is equal to 1 N m^{-2} . The equation for pressure is as follows:

$$P = \frac{F}{A}$$

Where:

- P is pressure
- F is force
- A is the area of the surface upon which the force is acting

The pressure due to the weight of the liquid a depth below the liquid surface is given by:

$$P = \rho gh$$

Where:

- ρ is the density of the liquid (This is a lowercase rho, NOT a w)
- g is the gravitational field strength (equivalent to acceleration due to gravity)
- h is the depth below the liquid surface

From the equation, we can see that pressure increases as depth increases.

However, this equation only accounts for the pressure due to the weight of the liquid, neglecting atmospheric pressure. A full equation accounting for both the liquid pressure and atmospheric pressure is as follows:

$$P = \rho gh + P_{atm}$$

Where:

- P_{atm} is the atmospheric pressure. At sea level on Earth, the atmospheric pressure is 101325 Pa

Other units of pressure include atm (Measuring pressure with reference to atmospheric pressure at sea level, 1 atm = 101325 Pa) and mmHg (Measuring pressure with reference to a column of mercury, 1 mmHg = 133.322 Pa).

For gasses, since their density is so low, the pressure due to the weight of the gas is generally negligible. Hence, the pressure in a container of gas can be considered to be uniform:

$$P = P_{ext}$$

Where:

- P_{ext} is the pressure due to external forces on the container

11.1.1. Hydraulic Press Systems

Pascal's principle states that changes to the pressure applied in an enclosed incompressible fluid is transmitted to every part of the fluid and the walls of its container without loss. This principle is utilised in the operation of hydraulic presses.

Consider container connected to two pistons with face areas A_1 and A_2 respectively. When a force F_1 is applied on the first piston, the resulting change in the pressure of the liquid, P , is $\frac{F_1}{A_1}$. Then, the resulting force acting on the second piston, F_2 , is $PA_2 = P \frac{A_2}{A_1}$.

As we can see, the force on the second piston is not equal to the force on the first piston. By increasing A_2 and decreasing A_1 , we can greatly increase the value of F_2 with changing F_1 . This allows hydraulic presses to exert extremely large forces using only a small driving force.

11.2 Buoyancy & Resistive Forces

11.2.1. Buoyancy

When an object is placed in a fluid, it displaces a volume, V , of the fluid. Due to the differences in the pressure at the top surface of the object and the pressure at the bottom surface of the object, a buoyant force is produced. The magnitude of the buoyant force is equivalent to the weight of fluid displaced:

$$F_B = \rho Vg$$

Where:

- F_B is the buoyant force
- ρ is the density of the fluid
- V is the volume of liquid displaced
- g is the gravitational field strength (equivalent to acceleration due to gravity)

11.2.2. Stokes's Law

When an object is moving in a fluid, it experiences a resistive force known as drag. At low speeds, the drag force acting on a spherical object is modelled by Stokes's Law:

$$F_D = -6\pi\eta r$$

Where:

- F_D is the drag force
- η is the dynamic viscosity of the fluid (unit: $\text{kg m}^{-1} \text{s}^{-1}$) (This is a lowercase eta, NOT an n)
- v is the velocity of the object

The minus symbol indicates that the drag force acts oppositely to the direction of motion.

For non-spherical objects, the drag force at low speeds is assumed to be directly proportional to the velocity of the object:

$$F_D = -bv$$

Where:

- b is a constant (unit: kg s^{-1})

Note that these equations only apply to low-speed flow.

11.3 Laminar & Turbulent Flow

Laminar flow is characterised by the flow of a fluid in parallel layers, without disruption and interaction between different layers.

Turbulent flow is characterized by the irregular flow of fluids in which there are inconsistent flow patterns and variations in velocities throughout the fluid.

11.3.1. Flow Tubes

For an incompressible, viscous fluid under laminar flow in a cylindrical tube, the volumetric flow rate is given by:

$$Q = \frac{\pi r^4 \Delta P}{8\eta L}$$

Where:

- Q is the volumetric flow rate
- r is the radius of the tube
- ΔP is the difference in pressure at the entrance and exit of the tube
- η is the dynamic viscosity
- L is the length of the pipe

11.3.2. Continuity Law

The continuity law states that the mass flow rate at any point in a pipe must be uniform, regardless of the geometry of the pipe. For incompressible fluids, this also implies that the volumetric flow rate is uniform.

Consider a fluid flowing through a pipe with a cross-sectional area of A_1 at its opening and a cross-sectional area of A_2 at its exit. Let the velocity of fluid entering the pipe be v_1 .

To calculate the volumetric flow rate at the entrance, Q_1 , we multiply velocity by the cross-sectional area:

$$Q_1 = v_1 A_1$$

From the continuity law, the volumetric flow rate at the exit, Q_2 , is equal to Q_1 :

$$Q_2 = Q_1$$

We can express the exit velocity of the fluid in terms of the cross-sectional areas and entrance velocity:

$$\begin{aligned} v_2 A_2 &= v_1 A_1 \\ v_2 &= v_1 \frac{A_1}{A_2} \end{aligned}$$

11.3.3. Bernoulli's Principle

Bernoulli's Principle states that the pressure within a fluid decreases as its speed increases.

11.3.4. Bernoulli's Equation

Bernoulli's equation can be derived from the conservation of energy. It states that the total energy density of a fluid remains constant along a streamline. This equation is generally only relevant to laminar flow.

Streamlines refer to the individual paths of fluid flow within a system. An important property of streamlines is that they do not intersect one another.

The total energy density of a fluid at any point can be taken as:

$$u = P + \frac{1}{2} \rho v^2 + \rho gh$$

Where:

- u is the total energy density
- P is pressure, representing the internal energy density of the fluid
- $\frac{1}{2}\rho v^2$ is the kinetic energy density of the fluid
- ρgh is the gravitational potential energy density of the fluid.

This equation is very useful when solving for the pressure or velocity of fluids.

11.4 Surface Tension

11.5 Capillary Pressure

12. Young's, Bulk, & Shear Modulus

All materials exhibit elasticity to some extent. When forces are applied on an object, it deforms. The extent of its deformation is defined by its various moduli.

However, stresses are often used instead of forces when performing calculations as they are easier to work with. Stress can be defined as the force applied per unit area. It has the unit Pa.

The deformation of the object is quantified as strain. Strain is the ratio between the change in the length of an object and its original length. Since it is a ratio, it is a dimensionless quantity.

$$\text{strain} = \frac{\Delta x}{x}$$

Where:

- x has the dimension of length
- Δx is the change in x

12.1.1. Young's Modulus

The Young's modulus, Y , is applicable when the object is experiencing tensile (pulling) stress.

$$Y = \frac{\text{stress}}{\text{strain}}$$

12.1.2. Bulk Modulus

The bulk modulus, B , is applicable when the object is experiencing hydrostatic stress (Force is directed into the object).

$$B = \frac{\text{stress}}{\text{strain}}$$

12.1.3. Shear Modulus

The shear modulus, G , is applicable when the object is experiencing shear (lateral) stress.

$$G = \frac{\text{stress}}{\text{strain}}$$

12.1.4. Calculations

To calculate the change in the dimension of an object, use the following equation:

$$\Delta x = x \frac{\text{stress}}{k}$$

Where:

- x is the original value of the dimension
- Δx is the change in the value of the dimension
- k is the applicable stress modulus (Y , B , or G)

In most cases, the Young's modulus and bulk modulus of an object are roughly equal.

Note that the modulus of a material usually does not remain constant as stress varies.

13. Gravity

13.1 Newton's Law of Gravity

Newton's law of gravity states that the force of gravitational attraction between two objects is given by the equation:

$$F = G \frac{m_1 m_2}{r^2}$$

Where:

- F is the gravitational force
- G is the universal gravitational constant ($6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$)
- m_1 is the mass of object 1
- m_2 is the mass of object 2
- r is the distance between the centres of mass of the 2 objects

The gravitational field strength due to an object at a distance can be calculated as:

$$g = G \frac{m}{r^2}$$

Where:

- g is the gravitational field strength
- G is the universal gravitational constant ($6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$)
- m is the mass of the object
- r is the distance between the centre of mass of the object and the point

Note that this only applies to spherical objects.

13.2 Newton's Shell Theorem

Newton's Shell Theorem states that the outside the radius of a hollow sphere, the gravitational field of the hollow sphere is equivalent to the gravitational effect of a point mass at the centre of the sphere with equal mass. Simply put, hollow spheres can be treated as point mass at their centres when calculating their gravitational effects. This also applies to solid spheres (Which can be treated as a series of nested hollow spheres).

Inside the hollow sphere, the gravitational field strength is equal to zero.

13.3 Gravitational Potential Energy

GPE can be calculated as the work done to bring two objects infinitely far from one another to a distance from each other. First, let's assume that object 1 has a far greater mass than object 2 (e.g. the Earth and a person). We assign the symbol M to the mass of object 1 and the symbol m to the mass of object 2. We use the symbol R to represent the distance between the two objects.

We will use a frame of reference where object 1 is stationary, hence only object 2 is moving. Moreover, we will use the value of GPE when the objects are infinitely far apart as the reference value for zero.

To find GPE, we integrate the gravitational force

$$\begin{aligned}
 GPE &= \int_{\infty}^R G \frac{Mm}{r^2} dr \\
 &= \left[-G \frac{Mm}{r} \right]_{\infty}^R \\
 &= -G \frac{Mm}{R} + G \frac{Mm}{\infty} \\
 &= -G \frac{Mm}{R}
 \end{aligned}$$

The minus symbol indicates that GPE decreases as the objects move closer to one another ($\frac{1}{R}$ increases; R decreases).

To find the change in GPE as object 2 moves from a first point to a second point, we can use the following equation:

$$\Delta GPE = -G \frac{Mm}{R_2} + G \frac{Mm}{R_1} = GMm \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Where:

- ΔGPE is the change in GPE
- M is the mass of object 1
- m is the mass of object 2
- R_2 is the distance between the objects at the second point
- R_1 is the distance between the objects at the first point

When the difference between R_1 and R_2 (which we will assign the symbol ΔR) is much smaller than the values of R_1 and R_2 , the gravitational force is approximately constant:

$$\begin{aligned} \Delta GPE &= -GMm \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ &= GMm \left(\frac{1}{R_1} - \frac{1}{R_1 + \Delta R} \right) \\ &= GMm \left(\frac{R_1 + \Delta R - R_1}{R_1(R_1 + \Delta R)} \right) \\ &= GMm \left(\frac{\Delta R}{R_1(R_1 + \Delta R)} \right) \end{aligned}$$

Since $R_1 \gg \Delta R$, $R_1 + \Delta R \approx R_1$:

$$\begin{aligned} GPE &\approx GMm \left(\frac{\Delta R}{R_1} \right) \\ GPE &\approx m \left(G \frac{M}{R_1} \right) \Delta R \end{aligned}$$

If we substitute $G \frac{M}{R_1}$ for g and substitute ΔR for h , the equation is identical to the simplified GPE equation from earlier.

13.4 Escape Velocity

The escape velocity of an object is the velocity it requires to escape the gravitational influence of another object. Imagine a spaceship of mass m orbiting a planetary body of mass M at a distance R . For the spaceship to escape the gravitational influence of the planetary body, they would have to be infinitely far from each other.

From the earlier derivation of GPE, we know that as the spaceship approaches the planetary body from infinity, the change in GPE is $-\frac{GMm}{R}$. Hence, for the spaceship to achieve infinite separation, it must have kinetic energy equivalent to $\frac{GMm}{R}$ (When all the KE is converted to GPE, the spaceship will be at infinity):

$$KE = -GPE$$

$$\frac{1}{2}mv^2 = \frac{GMm}{R}$$

$$v^2 = \frac{2GM}{R}$$

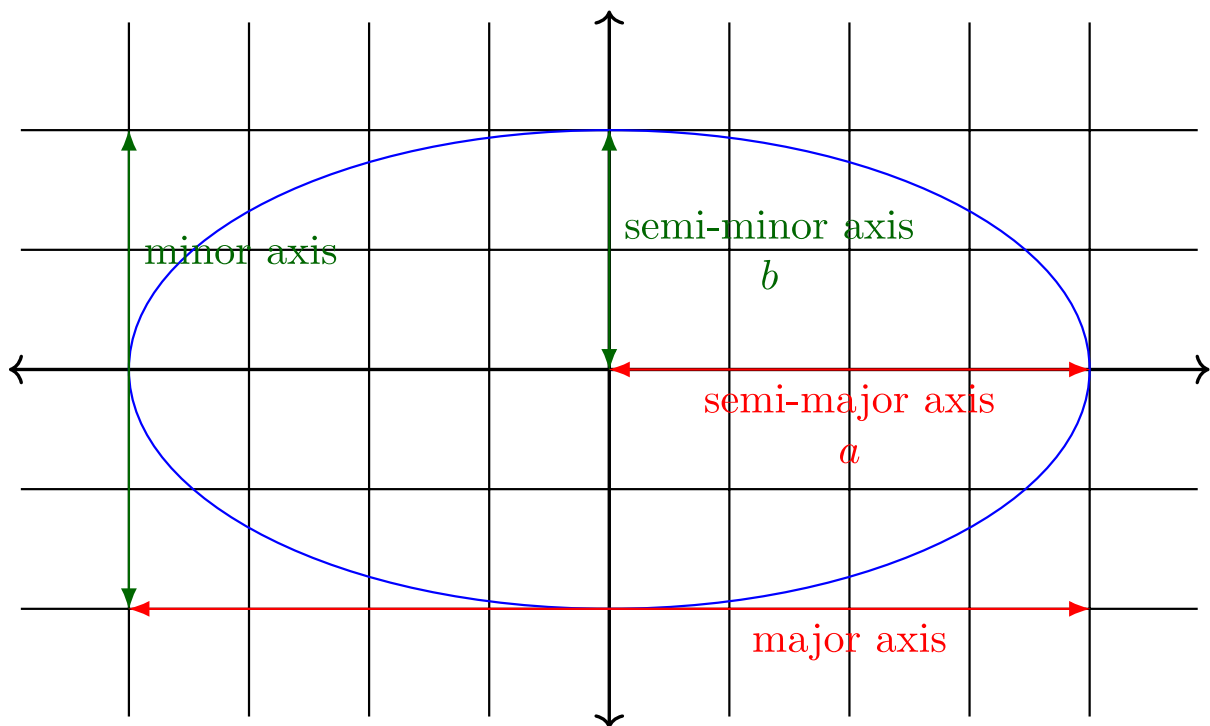
$$v = \sqrt{\frac{2GM}{R}}$$

Thus, the escape velocity is equal to $\sqrt{\frac{2GM}{R}}$

13.5 Kepler's Laws

13.5.1. Ellipses

Ellipses are a shape which resemble stretch circles.



Some important aspects of an ellipse are:

- **Major axis**
The major axis is defined as the maximum diameter of the ellipse.
- **Semi-major axis**
The semi-major axis is defined as half the major axis (i.e. the 'maximum radius' of the ellipse). A common symbol for the semi-major axis is a .
- **Minor axis**
The minor axis is defined as the minimum diameter of the ellipse.
- **Semi-minor axis**

The semi-minor axis is defined as half the minor axis (i.e. the 'minimum radius' of the ellipse)

- **Foci**

For any point on the edge of an ellipse, the sum of the distances from this edge to the two foci is a constant.

13.5.2. 1st Law

Kepler's 1st law states that the orbit of a planet is an ellipse with the Sun at one of the foci of the ellipse. More generally, for an object in orbit around a planetary body, its orbit follows an elliptical path with the planetary body at one of the foci of the ellipse.

13.5.3. 2nd Law

Kepler's 2nd law states that the sectorial velocity of an orbiting object is constant. This means that the area swept out by an imaginary line connecting the object to the planetary body per unit time is a constant.

Qualitatively, this means that as the distance between the orbiting object and the planetary body decreases, the velocity of the object increases.

13.5.4. 3rd Law

Kepler's 3rd law states that for circular orbits, the square of the orbital period of any object is directly proportional to the cube of its radius.

$$T^2 \propto r^3$$

Where:

- T is the time taken for one revolution about the main planetary body
- r is the radius of the orbit

Note that the mass of the orbiting bodies does not affect the orbital period. From this relation, we can relate the orbital periods and radii of two objects orbiting the same main planetary body.

$$\frac{T_1^2}{T_2^2} \propto \frac{r_1^3}{r_2^3}$$

Where:

- T_1 is the time taken for one revolution about the main planetary body for orbit 1
- r_1 is the radius of the orbit 1
- T_2 is the time taken for one revolution about the main planetary body for orbit 2
- r_2 is the radius of the orbit 2

14. Bibliography

Fundamentals of Physics (10th edition) Halliday, Resnick and Walker

<https://byjus.com/jee/moment-of-inertia/>

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